

练习 (不懂的地方再回去看视频)

求 First 练习

$A \rightarrow aB | \epsilon$

$A \rightarrow c$

First(A)=__

$A \rightarrow Ba$

$B \rightarrow b$

First(A)=__

$A \rightarrow Bc$

$B \rightarrow b | \epsilon$

First(A)=__

$A \rightarrow BC$

$B \rightarrow b | \epsilon$

$C \rightarrow c | \epsilon$

First(A)=__

例题

经消去直接左递归后变成: $E \rightarrow TE'$

(G(E))

$E' \rightarrow + TE' | \epsilon$

$T \rightarrow FT'$

(4.2)

$T' \rightarrow * FT' | \epsilon$

$F \rightarrow (E) | i$

例 4.7 对于文法(4.2), 我们可构造出每个非终结符的 FIRST 和 FOLLOW 集合:

FIRST(E)

FOLLOW(E)

FIRST(E')

FOLLOW(E')

FIRST(T)

FOLLOW(T)

FIRST(T')

FOLLOW(T')

FIRST(F)

FOLLOW(F)

习题 1

注意: [if] 是一个终结符, 同理[b][other][else][then]

```
1 | G(S): S->IETSP | 0
2 | I->if
3 | E->b
4 | O->other
5 | L->else
6 | T->then
7 | P->LS | ε
```

First	Follow
First(S)	Follow(S)
First(I)	Follow(I):
First(E)	Follow(E)
First(O):	Follow(O)
First(L)	Follow(L)
First(P)	Follow(P)
First(T):	Follow(T)

习题 2

```
1 | G(E): E->TE'
2 | E' -> +E | ε
3 | T->FT'
4 | T' -> T | ε
5 | F->PF'
6 | F' -> *F' | ε
7 | P->(E) | a | b | ^
```

First	Follow
First(E)	Follow(E):
First(E')=	Follow(E'):
First(T)=	Follow(T):
First(T')=	Follow(T')=
First(F)	Follow(F)=
First(F')	Follow(F')
First(P)=	Follow(P)=

练习答案 (做完再对答案)

$A \rightarrow aB | \epsilon$

$A \rightarrow c$

$\text{First}(A) = \{a, \epsilon, c\}$

$A \rightarrow Ba$

$B \rightarrow b$

$\text{First}(A) = \{b\}$

$A \rightarrow Bc$

$B \rightarrow b | \epsilon$

$\text{First}(A) = \{b, c\}$

$A \rightarrow BC$

$B \rightarrow b | \epsilon$

$C \rightarrow c | \epsilon$

$\text{First}(A) = \{b, c, \epsilon\}$

例题

经消去直接左递归后变成: $E \rightarrow TE'$

G(E):

$$\begin{aligned} E' &\rightarrow + TE' | \epsilon \\ T &\rightarrow FT' \\ T' &\rightarrow * FT' | \epsilon \\ F &\rightarrow (E) | i \end{aligned}$$

(4.2)

例 4.7 对于文法(4.2), 我们可构造出每个非终结符的 FIRST 和 FOLLOW 集合:

$\text{FIRST}(E) = \{ (, i \}$	$\text{FOLLOW}(E) = \{), \# \}$
$\text{FIRST}(E') = \{ +, \epsilon \}$	$\text{FOLLOW}(E') = \{), \# \}$
$\text{FIRST}(T) = \{ (, i \}$	$\text{FOLLOW}(T) = \{ +,), \# \}$
$\text{FIRST}(T') = \{ *, \epsilon \}$	$\text{FOLLOW}(T') = \{ +,), \# \}$
$\text{FIRST}(F) = \{ (, i \}$	$\text{FOLLOW}(F) = \{ *, +,), \# \}$

习题 1

注意：[if] 是一个终结符，同理[b][other][else][then]

```
1 G(S) : S->IETSP | 0
2 I->if
3 E->b
4 O->other
5 L->else
6 T->then
7 P->LS | ε
```

First	Follow
First(S)={if,other}	Follow(S)={#,else}
First(I)={if}	Follow(I)={b}
First(E)={b}	Follow(E)={then}
First(O)={other}	Follow(O)={else,#}
First(L)={else}	Follow(L)={if,other}
First(P)={else,ε}	Follow(P)={else,#}
First(T)={then}	Follow(T)={if,other}

习题 2

```
1 G(E) : E->TE '
2 E ' ->+E | ε
3 T->FT '
4 T ' ->T | ε
5 F->PF '
6 F ' ->*F ' | ε
7 P->( E ) | a | b | ^
```

First	Follow
First(E)={(,a,b,^}	Follow(E)={#,)}
First(E')={+,ε}	Follow(E')={#,)}
First(T)={(,a,b,^}	Follow(T)={+,#,)}
First(T')={(,a,b,^,ε}	Follow(T')={+,#,)}
First(F)={(,a,b,^}	Follow(F)={(,a,b,^,+,#,)}
First(F')={*,ε}	Follow(F')={(,a,b,^,+,#,)}
First(P)={(,a,b,^}	Follow(P)={*,(,a,b,^,+,#,)}